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CAUSALITY ≠ INVARIANCE: FUNCTION AND CONCEPT VECTORS IN LLMS

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ABSTRACT

Do large language models (LLMs) represent concepts abstractly, i.e., independent of input format? We revisit Function Vectors ($\mathcal{FV}$s), compact representations of in-context learning (ICL) tasks that causally drive task performance. Across multiple LLMs, we show that $\mathcal{FV}$s are not fully invariant: $\mathcal{FV}$s of the same concept are nearly orthogonal when extracted from different input formats (e.g., open-ended vs. multiple-choice). We identify Concept Vectors ($\mathcal{CV}$s), which carry more stable concept representations. Like $\mathcal{FV}$s, $\mathcal{CV}$s are composed of attention head outputs; however, unlike $\mathcal{FV}$s, the constituent heads are selected using Representational Similarity Analysis (RSA) based on whether they encode concepts consistently across input formats. While these heads emerge in similar layers to $\mathcal{FV}$-related heads, the two sets are largely distinct, suggesting different underlying mechanisms. Steering experiments reveal that $\mathcal{FV}$s excel in-distribution, when extraction and application formats match (e.g., both open-ended in English), while $\mathcal{CV}$s generalize better out-of-distribution across both question types (open-ended vs. multiple-choice) and languages. Our results show that LLMs do contain abstract concept representations, but these differ from those that drive ICL performance.

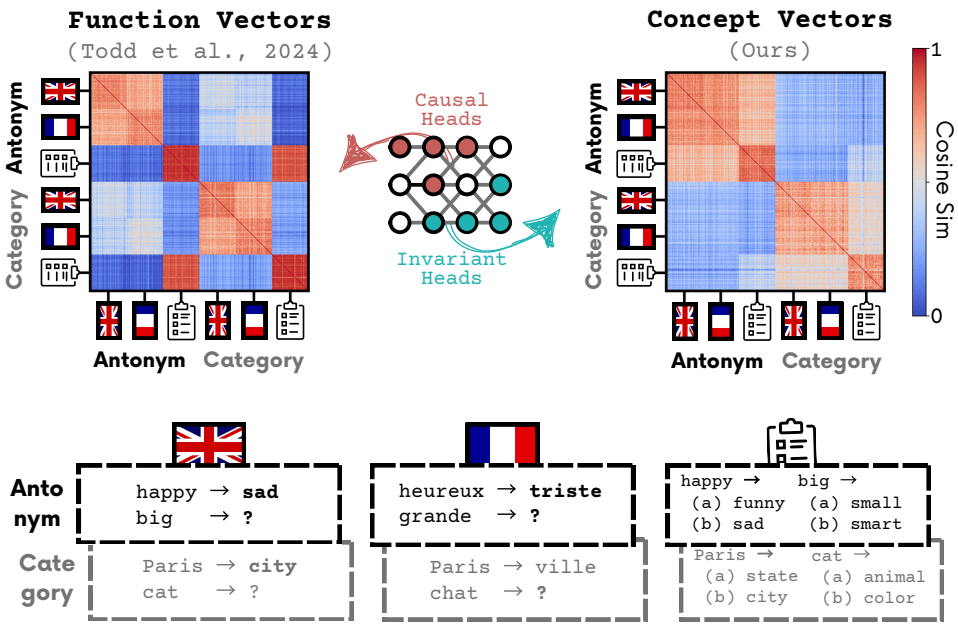

Figure 1: *Function vs. Concept Vectors*. **Top**: Similarity matrices for $\mathcal{FV}$s (left) and $\mathcal{CV}$s (right) in Llama 3.1 70B; cells show how similar two prompt representations are (warmer = more similar). **Middle**: Schematic highlighting the distinction between heads with causal effect (Activation Patching-selected) and heads that encode format-invariant structure (RSA-selected). **Bottom**: Example prompts for two concepts across three formats (EN open-ended, FR open-ended, multiple-choice). **Takeaway**: $\mathcal{FV}$s cluster by input format; $\mathcal{CV}$s cluster by concept across formats.

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1 INTRODUCTION

Do large language models represent concepts abstractly, i.e., in a way that is stable across surface form? We focus on *relational concepts*: mappings between entities, such as linking a word to its antonym. Cognitive science has long argued that abstract representation of such structure underlies human generalization (Gentner, 1983; Hofstadter, 1995; Mitchell, 2020). This capability allows identifying, by analogy, that "hot → cold" and "big → small" share the same oppositional relation, independent of the specific words or how the task is presented. Recent work shows that LLMs exhibit representational structures similar to humans (Pinier et al., 2025; Du et al., 2025; Doerig et al., 2025), raising the question: *do the abstract representations hypothesized to support analogical reasoning actually drive LLM performance on such tasks?*

We find that LLMs do contain abstract relational concept information, but the components that encode it differ from those that causally drive in-context learning (ICL) behavior. This separation challenges the *single-circuit hypothesis* that format-invariant representations are what primarily enable ICL.

We revisit Function Vectors ($\mathcal{FV}$s)—compact vectors formed by summing outputs of a small set of attention heads that mediate ICL (Todd et al., 2024; Hendel et al., 2023; Yin & Steinhardt, 2025). Because $\mathcal{FV}$s transfer across contexts (e.g., differently formatted prompts and natural text), they are often treated as encoding the underlying concept (Zheng et al., 2024; Griffiths et al., 2025; Bakalova et al., 2025; Brumley et al., 2024; Fu, 2025). We update this view: $\mathcal{FV}$s are not fully invariant. For the same concept, $\mathcal{FV}$s extracted from different input formats (open-ended vs. multiple-choice) are nearly orthogonal, indicating that $\mathcal{FV}$s mix concept with format (§2.2.1).

To isolate format-invariant structure, we contrast *activation patching* (AP), which localizes components with causal effects on outputs, with *representational similarity analysis* (RSA) (Kriegeskorte, 2008), which localizes components whose representations organize by concept independent of format. Using RSA to select heads and then summing their activations yields Concept Vectors ($\mathcal{CV}$s). Across seven relational concepts, three input formats (open-ended English, open-ended French, multiple-choice), and four models (Llama 3.1 8B/70B; Qwen 2.5 7B/72B), we find that $\mathcal{CV}$ heads arise in similar layers but are largely disjoint from $\mathcal{FV}$ heads, suggesting separable mechanisms for invariance vs. causality (§2.2.2).

Finally, we test whether $\mathcal{CV}$s can steer. In steering experiments, $\mathcal{FV}$s produce larger in-distribution gains when extraction and application formats match (§3.2.1), whereas $\mathcal{CV}$s generalize more consistently out-of-distribution across question type and language (§3.2.2) and produce fewer format artifacts (e.g., tokens and language from extraction prompts; §3.2.3).

Overall, our contributions are as follows:

- **$\mathcal{FV}$s are not input-invariant.** They mix relational concepts with input format; same-concept $\mathcal{FV}$s differ sharply across formats.
- **RSA reveals $\mathcal{CV}$ heads.** These heads encode relational concepts at a higher level of abstraction than $\mathcal{FV}$ heads.[1]
- **$\mathcal{CV}$ and $\mathcal{FV}$ heads are disjoint.** $\mathcal{FV}$s and $\mathcal{CV}$s are realized by different attention heads, suggesting that abstract concept representations are distinct from the mechanisms that causally drive ICL performance.
- **Steering trade-off.** $\mathcal{FV}$s steer more strongly in-distribution, while $\mathcal{CV}$s generalize more consistently out-of-distribution, albeit with smaller absolute gains.

2 IN SEARCH OF INVARIANCE

We test whether concept representations are stable across surface form, using AP (causal heads) and RSA (format-invariant heads) across models, datasets, and formats. We then form Function/Concept Vectors to compare clustering by format vs. concept; AP/RSA heads lie in similar layers but show minimal top-K overlap.

[1]We expand on what we mean by "higher level of abstraction" in the Discussion (§5).

2.1 METHODS

2.1.1 MODELS

We test Llama 3.1 (8B, 70B) and Qwen 2.5 (7B, 72B) models (Meta AI, 2024; Qwen et al., 2025). All models are autoregressive, residual-based transformers (Vaswani et al., 2023). Each model, f internally comprises of $\mathcal{L}$ layers. Each layer is composed of a multi-layer perceptron (MLP) and J attention heads $a_{\ell j}$ which together produce the vector representation of the last token of layer ℓ, $\mathbf{h}_\ell = \mathbf{h}_{\ell-1} + \text{MLP}_\ell + \sum_{j \in J} a_{\ell j}$ (Elhage et al., 2021).

2.1.2 TASKS

Datasets We define a dataset as one concept expressed in one input format (e.g., Antonym in open-ended English). For each dataset we build a set of in-context prompts $P_d = \{p_d^i\}$ where i indexes individual prompts within dataset d. Each prompt contains few-shot input–output examples (x, y) that illustrate the same concept, followed by a query input x_q^i whose target output y_q^i is withheld. The input–output pairs (x, y) were either sourced from prior work or generated using OpenAI's GPT-4o (see Appendix D for details). Example prompts are provided in Appendix A.

Concepts. We consider seven concepts:

- **Antonym** Map a word to one with opposite meaning (e.g., hot → cold).
- **Categorical** Map a word to its semantic category (e.g., apple → fruit).
- **Causal** Map a cause to an effect (e.g., rain → wet).
- **Synonym** Map a word to one with similar meaning (e.g., big → large).
- **Translation** Translate a word to another language (e.g., house → maison).
- **Present–Past** Convert a verb from present to past tense (e.g., run → ran).
- **Singular–Plural** Convert a noun from singular to plural (e.g., cat → cats).

Input formats. We vary only the prompt's surface format; the (x, y) relation stays the same. Formats:

- Open-ended ICL in English (`OE-EN`)
- Open-ended ICL in a different language (French or Spanish; `OE-FR` or `OE-ES`)
- Multiple-choice ICL in English (`MC`)

We use 5-shot prompts for open-ended and 3-shot for multiple-choice to reduce computational load given prompt length. Altogether, we have 21 datasets (7 concepts × 3 input formats). We build 50 prompts per dataset (total $N = 1050$ prompts).

2.1.3 ACTIVATION PATCHING

Activation patching replaces specific activations with cached ones from a *clean* run to assess their impact on the model's output. The cached activations are then inserted into selected model components in a *corrupted* run, where the systematic relationships in the prompt are disrupted. For example, in an antonym ICL task, consider a *clean prompt*: `Hot → **Cold**, Big → **Small**, Clean → **?**` and a *corrupted prompt*: `House → Cold, Eagle → Small, Clean → **?**` The goal of activation patching is then to localize model components that push the model to the correct answer, **`Dirty`**, on the corrupted prompt.

We compute the *causal indirect effect* (CIE) for each attention head $a_{\ell j}$ as the difference between the probability of predicting the expected token y when processing the corrupted prompt $\tilde{p}$ with and without the transplanted mean activation $\bar{\mathbf{a}}_{\ell j}$ from clean runs:

$$\text{CIE}(a_{\ell j}) = f\Big(\tilde{p} \mid \mathbf{a}_{\ell j} := \bar{\mathbf{a}}_{\ell j}\Big)[y] - f(\tilde{p})[y] \tag{1}$$

We then compute the *average indirect effect* (AIE) over a collection $\mathcal{D}$ of all datasets (§2.1.2). [2]

$$\text{AIE}(a_{\ell j}) = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \frac{1}{|\tilde{\mathcal{P}}_d|} \sum_{\tilde{p}_i \in \tilde{\mathcal{P}}_d} \text{CIE}(a_{\ell j}) \tag{2}$$

where $\tilde{\mathcal{P}}_d$ denotes the set of corrupted prompts for dataset d.

[2] *Note*: Unlike Todd et al. (2024) we compute AIE scores across all input formats, not `OE-ENG` only.